DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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***Abstract*— Physics-based humanoid imitation requires a controller to jointly match heterogeneous objectives involving global motion, articulated pose, and velocity. Existing tracking pipelines commonly combine these objectives through manually designed reward terms and motion-dependent coefficients, so a controller that tracks one skill well may need a different reward for the next. We introduce DARE, a differential adversarial reward that learns this aggregation directly from synchronized reference–policy errors. DARE preserves the semantic factorization of the differential before learning cross-objective interactions, so root, pose, and velocity residuals enter the learned reward through a common structured interface. Because perfect imitation is the fixed zero differential, DARE controls the critic through its score change relative to this unique anchor, yielding a globally bounded reward geometry trained by binary adversarial supervision. Using one frozen reward-learning configuration, DARE tracks six heterogeneous skills with position errors from $0.021\,\mathrm{m}$ to $0.118\,\mathrm{m}$ and DoF-velocity errors from $0.271\,\mathrm{rad/s}$ to $1.012\,\mathrm{rad/s}$, without manually weighted tracking rewards or per-motion reward tuning.**

I. Introduction

Physics-based humanoid control must reproduce a prescribed reference motion under dynamics, actuation limits, and intermittent contact. Whether an imitation is accurate cannot be judged from a single geometric residual. Global root displacement and orientation, articulated body configuration, joint velocity, momentum, and contact transitions all have to be satisfied together. These objectives interact: a small mismatch in contact timing can change which poses and velocities remain physically attainable a few steps later. On long-horizon, contact-rich skills that coupling is especially strong, so the relevant trade-off among objectives is nonlinear in the tracking residual.

The standard engineering response is to define a tracking term for each error and combine those terms by hand. A representative scalar reward is

$$r_t^{\text{hand}} = \sum_{g=1}^{G} w_g \exp\bigl[-\alpha_g e_g(s_t, \hat{s}_t)\bigr]. \tag{1}$$

The weights w_g decide how objectives exchange against one another, and the scales α_g fix the numerical range of each error. Implementations often add joint-specific weights, contact schedules, alive bonuses, or clip-specific termination rules, each of which further couples learned behavior to a manual scalarization. When the motion changes, typical error magnitudes, learning difficulty, and dynamical roles all change, so the same coefficients need not retain the same meaning. A setting that tracks a cyclic walk can starve the brief contact sequence of a vault, while a setting tuned for an aerial skill can over-penalize the long ground phase of a get-up. Reward construction itself becomes a substantial part of controller engineering.

Data-driven adversarial rewards offer a different route. Instead of declaring how many meters of root error are worth how many radians of joint error, a discriminator can learn a scalar imitation signal from the difference between reference and policy behavior. Style priors such as AMP do this at the level of short state transitions [13]. Adversarial differential discriminators (ADD) make the tracking case precise: imitation is a decision between an ideal residual and the residuals produced by the current policy [17]. Mapping every perfect pair to the same origin removes the need to invent a success threshold per feature, and it removes the need to write (1). Completing a new skill, however, still typically requires retuning how that discriminator itself is trained. A regularizer that stabilizes one clip can be too strong or too weak for another, so the engineering burden moves from tracking kernels to critic coefficients. We take learned aggregation as the right direction, and ask for a reward model that can be frozen once and reused.

Unlike a generic observation, a motion differential has two structural facts that a reward model can use. Perfect imitation is always the same point, $\Delta = 0$. Actual residuals occupy semantic subspaces associated with root motion, body configuration, rotation, and velocity. These subspaces are the coordinates in which tracking quality is already measured. If the first nonlinear layer mixes all of them, later aggregation is exposed to raw input dimension: a high-dimensional body-pose block can dominate a three-dimensional root-position block simply because it offers more coordinates, not because it is more important for the skill. DARE therefore factorizes before interaction. Each differential factor is mapped to an equal-capacity representation; a shared critic then learns the full nonlinear coupling among objectives.

The second fact yields a stability condition that does not require a motion-specific penalty coefficient. Because every perfect pair occupies the same origin, the critic's meaningful variation is the change in score from that origin to a realized residual. A global gain bound on this anchor-relative change keeps nearby residuals from producing disproportionate reward jumps, while bias terms and a shared nonlinear trunk remain free to express the geometry of different skills. We realize that geometry by spectrally parameterizing the equalized critic, and we reuse those maps for every motion.

The contributions are as follows. First, we formulate humanoid motion imitation as learning a nonlinear scalar reward over semantically factorized motion differentials, eliminating manually weighted tracking objectives. Second, we introduce DARE, an equalized differential critic that preserves each semantic factor before interaction and controls reward variation relative to the unique zero-error anchor. Third, we evaluate DARE with one frozen reward-learning configuration across six heterogeneous skills—Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick—and show accurate tracking without per-motion reward tuning.

II. Related Work

A. Physics-Based Humanoid Motion Imitation

DeepMimic showed that reinforcement learning can reproduce diverse motion-capture clips with physically simulated characters by combining pose, velocity, root, and end-effector tracking terms [1]. Each term is an exponential kernel of a physical error, and the kernels are summed with coefficients chosen by the user. Subsequent work scales tracking to large motion collections, improves recovery, and learns more general controllers through architecture and data [2]–[7]. Distributed RL for humanoids demonstrated that physically simulated characters can acquire locomotion from weak task rewards [8], and SFV showed that reference tracking can be driven from video [9]. Architecture-focused systems then scaled the controller: ScaDiver uses a mixture of experts, UniCon separates motion scheduling from low-level execution, PULSE distills a reusable latent motion representation, and MaskedMimic conditions control on incomplete goals. Motion matching and residual tracking controllers similarly keep a phase or error coordinate in the loop [10], [11]. These systems deliver strong control. Their imitation objectives, however, still commonly include manual scales and weights. The present concern is therefore an imitation objective that can be reused across motions.

B. Adversarial Imitation and Learned Rewards

GAIL casts imitation as adversarial occupancy matching [12]. AMP adapts this idea to physics-based characters by discriminating short state transitions and using the learned signal as a motion prior [13]. ASE extends the same prior to reusable skill representations [14]. Reward-learning variants such as AIRL seek discriminators whose scores remain meaningful under changes of dynamics [15], while Wasserstein critics replace Jensen–Shannon training with a Lipschitz value function [16]. Distribution matching is well suited to stylized behavior, where the policy need not be synchronized with a particular frame of a clip.

Frame-synchronized tracking is a stricter problem. A reward that only scores "looking like the dataset" can accept a plausible but incomplete motion. Differential rewards address this by handing the reference–policy error itself to a discriminator [17]. ADD composes multiple tracking objectives into one differential and treats the ideal zero vector as the sole positive example, thereby learning a nonlinear aggregation without component weights. Completing a new skill still typically requires retuning the discriminator's training coefficients. DARE continues the learned-aggregation program. Its emphasis is the representation of the differential before interaction, and a reward geometry that can be shared across motions.

C. Stable Adversarial Learning

Unconstrained adversarial critics can become overly sensitive. A discriminator that assigns a huge slope near the decision boundary produces brittle rewards, while a discriminator that saturates yields no learning signal. Gradient penalties, weight normalization, Lipschitz constraints, and spectral parameterization were introduced to control that sensitivity [18]–[20]. In motion tracking the same tools are often applied as an extra loss whose coefficient then varies with the clip. Once the ideal residual is a known origin, that concern has a natural form: nearby residuals should not produce arbitrarily different scores. DARE builds this geometry into an equalized critic, so one Lipschitz construction can serve every skill.

D. Multi-Objective Optimization

Weighted-sum scalarization, Pareto methods, adaptive loss balancing, and gradient-based multi-task optimization all address competing objectives [21]–[23]. They typically require explicit preferences, multiple objective gradients, or a Pareto set. Linear scalarization is simple, but it conceals weak objectives and forces the user to choose trade-offs before seeing the residual distribution of a learned policy.

The tracking errors of motion imitation form a multi-objective problem of this kind: root position, body rotation, and velocity cannot be optimized in isolation, and their relative urgency changes through a clip. Differential adversarial learning provides a data-driven nonlinear scalarization of that vector. DARE performs the aggregation with one learned critic and uses the same training configuration for different motions, rather than maintaining a preference vector or a family of task gradients.

III. Background

A. Reference-Conditioned Humanoid Control

Let s_t be the simulated state at time t, $\hat{s}_t$ the synchronized reference target, and a_t the action. The simulated state contains root pose and velocity, joint configuration and velocity, and body transforms. The action is a vector of desired joint positions. A proportional–derivative servo converts those targets into torques at the control frequency. The transition kernel p and discount $\gamma \in (0, 1)$ define a discounted MDP. A policy π_φ observes proprioception together with a short-horizon reference target—the next few synchronized frames—and must close the loop under contact and underactuation. Training begins from a randomized reference phase, so a single policy has to recover the clip rather than replay a fixed open-loop sequence. The objective is

$$J(\pi_\varphi) = \mathbb{E}_{\pi_\varphi}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right]. \tag{2}$$

The actor interface and PPO update follow the standard reference-conditioned setup [24], [25]. Section IV constructs the DARE reward on top of this interface and does not change the action space, the PD plant, or the PPO surrogate.

B. Motion Tracking as Multi-Objective Optimization

A motion descriptor decomposes a state into G feature groups $g_1(s), \ldots, g_G(s)$ covering global root motion, articulated body configuration, and velocities. For the g-th group, an error map E_g compares the simulated features with the synchronized reference and returns a scalar $e_{g,t}$—a distance in that group's units. Collecting these G scalars yields the tracking-error vector

$$\mathbf{e}_t = [e_{1,t}, \ldots, e_{G,t}], \qquad e_{g,t} = E_g\big(g_g(s_t), g_g(\hat{s}_t)\big). \quad (3)$$

Equation (1) is then a hand-chosen map from $\mathbf{e}_t$ to a reward: each E_g has already discarded direction, and the coefficients w_g and α_g decide the remaining trade-off. The difficulty is not only that G is larger than one. Root position is measured in meters, orientations in an embedding of rotation, and velocities in units per second; their errors change at different rates through a motion. The objectives also interact: a contact error changes which pose and velocity errors remain physically attainable, so the feasible set is not a product of independent intervals.

From a multi-objective view, the policy does not solve G independent tasks. It needs one scalar training signal that coordinates a single action sequence. A fixed weighted sum freezes the trade-off in advance and therefore encodes a preference that may be wrong for a later clip, or even for a later phase of the same clip. The remainder of the paper studies a learned scalarization of $\mathbf{e}_t$, so that the trade-off is inferred from residual data rather than declared before training.

C. Learned Motion Objectives

An adversarially learned objective distinguishes ideal or reference behavior from policy behavior and then trains the policy to maximize that signal. When the discriminator observes states or short transitions, it can reward plausible motion that is not synchronized with the clip. For synchronized imitation, a differential retains information about *how* the current motion departs from the reference—which bodies lead or lag, and which velocities disagree—rather than discarding that structure into a state occupancy. Perfect tracking collapses every pair onto one origin, so the positive class does not have to be a collection of privileged frames.

We seek a single learned reward that (i) aggregates heterogeneous motion errors without manually specified component weights, (ii) preserves their semantic structure before learning interactions, and (iii) remains stable under one shared training configuration. Section IV derives DARE from these requirements.

IV. DARE: Differential Adversarial Reward Equalization

A. Learned Differential Reward

The distances $e_{g,t}$ are convenient for a weighted sum, but they throw away the signed structure of the residual. Two policies can share the same root-position distance while drifting in opposite directions; only one of them remains compatible with the next contact. We therefore work with the signed, normalized motion differential.

Recall that s_t is the simulated state and $\hat{s}_t$ the synchronized reference. For each objective g, a feature map ϕ_g extracts the corresponding coordinates from a state. The concatenated map is $\phi(s) = \phi_1(s) \oplus \cdots \oplus \phi_G(s)$. A running, origin-preserving normalizer $\mathcal{N} = \mathcal{N}_1 \oplus \cdots \oplus \mathcal{N}_G$ rescales each coordinate by a magnitude estimated from training data, removing unit disparity without imposing a trade-off among groups. Quaternion orientations are mapped to a continuous tangent–normal embedding before subtraction, so the difference is taken in a Euclidean feature space. The group-wise residual, the full differential, and the ideal point are

$$\begin{aligned} \Delta_{g,t} &= \mathcal{N}_g\big(\phi_g(\hat{s}_t) - \phi_g(s_t)\big), \\ \Delta_t &= \Delta_{1,t} \oplus \cdots \oplus \Delta_{G,t}, \\ \Delta^+ &= 0. \end{aligned} \quad (4)$$

Equivalently, $\Delta_t = \mathcal{N}\big(\phi(\hat{s}_t) - \phi(s_t)\big)$. The vector Δ_t retains error direction and cross-coordinate relationships instead of collapsing them into G distances. Every perfectly synchronized pair maps to the same origin. Differentials from the current rollout and from a replay buffer form a non-ideal distribution q_π. Two characters at the same pose can still require different actions if their momenta disagree, so ϕ includes both configuration and velocity. Global root coordinates keep the displacement needed for travel and vaulting; root-relative body positions describe the articulated shape without copying the same translation onto every body. Each block corresponds to a physical tracking question that a handcrafted reward would have named explicitly.

The remaining question is how to turn Δ into a scalar that a policy can maximize. Rather than assigning a price to each block, we ask a critic to recognize the unique ideal point. Let $D_\theta(\Delta) = \sigma\big(f_\theta(\Delta)\big)$ be the probability that a residual is ideal, where σ is the logistic sigmoid and f_θ is a scalar logit. The origin is the positive class; residuals drawn from q_π are the negative class. The corresponding two-player objective, without an extra gradient penalty, is

$$\max_\theta \; \log D_\theta(0) + \mathbb{E}_{\Delta\sim q_\pi}\big[\log\big(1 - D_\theta(\Delta)\big)\big].$$

The first term asks the critic to accept perfect imitation; the second asks it to reject the residuals of the current controller. Because every perfect pair occupies the same origin, the positive class is a point rather than a set of "good enough" frames. Taking the expectation over policy residuals, as above, makes the negative class track both the present controller and recently visited errors in replay.

Minimizing the balanced Bernoulli negative log-likelihood of the same labeling is equivalent. Using $-\log\sigma(z) = \mathrm{softplus}(-z)$ and $-\log\big(1-\sigma(z)\big) = \mathrm{softplus}(z)$ yields the training loss

$$\boxed{\begin{aligned} \mathcal{L}_{\mathrm{DARE}}(\theta) = \tfrac{1}{2}\,&\mathrm{softplus}\big[-f_\theta(0)\big] \\ &+ \tfrac{1}{2}\mathbb{E}_{\Delta\sim q_\pi}\,\mathrm{softplus}\big[f_\theta(\Delta)\big]. \end{aligned}} \quad (5)$$

As the policy improves, q_π concentrates, and f_θ re-estimates how root, pose, and velocity residuals should trade. The aggregation is therefore a function of the residual distribution, not of a coefficient vector chosen before training. Equation (5) replaces the manual scalarization (1) by a data-driven nonlinear scalarization.

B. Equalized Anchor-Relative Critic

If the first nonlinear layer mixes all coordinates of Δ, later layers never see which residual asked which physical question. High-dimensional blocks—body pose, in particular—then dominate low-dimensional ones by coordinate count. The learned prices become an artifact of the feature layout: a skill that hinges on a three-dimensional root residual can still be scored as if body-pose coordinates were the cheaper currency, simply because there are more of them. Section V confirms the cost of that mixing. Flattening the first layer, while retaining a Lipschitz trunk of the same capacity, raises tracking error on representative skills. DARE therefore maps each factor independently to an equal-width latent before any shared mixing.

The differential space is the semantic direct sum $X = X_1 \oplus \cdots \oplus X_G$. The Differential Direct-Sum Layer applies an independent affine map and nonlinearity to each factor and concatenates equal-width latents:

$$\Phi_\oplus(\Delta) = \bigoplus_{g=1}^{G} \rho(\bar{W}_g \Delta_g + b_g), \qquad \bar{W}_g = W_g / \sigma_{\max}(W_g). \quad (6)$$

$$f_\theta(\Delta) = \bar{T}\big(\Phi_\oplus(\Delta)\big), \qquad \mathrm{Lip}(\bar{T}) \le 1. \quad (7)$$

Here ρ is a 1-Lipschitz activation and $\sigma_{\max}$ is the largest singular value. Each factor map includes an affine bias, so activation thresholds can sit where the residual distribution actually lives; spectral normalization of W_g keeps the map non-expansive regardless of that offset. Equalization then gives every motion factor the same latent capacity before interaction, so a three-dimensional root residual is not structurally outvoted by a much wider body-pose residual. The concatenated features enter one shared critic that can represent contact–pose–velocity couplings: an orientation error that is harmless at one angular velocity can be incompatible at another, and that dependence is not written as a product of kernels. The factors themselves are the natural tracking blocks of a humanoid differential: root position and rotation, body position and rotation, root linear and angular velocity, and joint velocity.

All linear maps, including the shared trunk $\bar{T}$, use spectral parameterization. Each block map $\phi_g(\Delta_g) = \rho(\bar{W}_g \Delta_g + b_g)$ is therefore non-expansive—the bias cancels in the Lipschitz argument—and the direct-sum layer satisfies

$$\begin{aligned}\big\|\Phi_\oplus(\Delta) - \Phi_\oplus(\Delta')\big\|_2^2 &= \sum_{g=1}^{G} \big\|\phi_g(\Delta_g) - \phi_g(\Delta'_g)\big\|_2^2 \\ &\le \|\Delta - \Delta'\|_2^2.\end{aligned} \quad (8)$$

The inequality is immediate: each spectrally normalized block has Lipschitz constant at most one, and the Euclidean norm of a concatenation is the root-sum of squared block norms. Equivalently, the Jacobian of $\Phi_\oplus$ is block diagonal across factors, so a perturbation in one semantic subspace cannot be amplified by the first layer through coordinates of another subspace. Mixing is deferred to $\bar{T}$, where every factor already occupies an equal-width channel. Equalization assigns every factor the same latent width before mixing; the subsequent trunk can then learn contact–pose–velocity couplings without the first layer converting coordinate cardinality into an implicit price.

Because every ideal pair occupies the same origin, the quantity that governs reward variation is the zero-anchor relative gap

$$G_\theta(\Delta) = f_\theta(0) - f_\theta(\Delta). \quad (9)$$

Combining (7) and (8) gives

$$\boxed{|G_\theta(\Delta)| = \big|f_\theta(0) - f_\theta(\Delta)\big| \le \|\Delta\|_2.} \quad (10)$$

If the realized motion changes only slightly relative to the reference, the learned score cannot jump by an amount larger than the residual itself. As the error distribution evolves, the shared nonlinear trunk can still learn the combinations required by different motions. Affine biases are retained throughout, so activation boundaries can move across residual regions and the reward geometry remains nonlinear inside the gain bound.

Differential equalization preserves the objectives before interaction; anchor-relative gain control makes their learned aggregation stable under a shared configuration.

C. DARE Reward and Training

The critic D_θ is a probability of ideality. A policy that maximizes imitation should therefore receive a large reward precisely when that probability is large, i.e., when $1 - D_\theta(\Delta)$ is small. The AMP-style conversion $-2\log\big(1 - D_\theta(\Delta)\big)$ implements this requirement and, in logits, is a scaled softplus:

$$\boxed{\begin{aligned} r_t^{\mathrm{DARE}} &= -2\log\big[1 - D_\theta(\Delta_t)\big] \\ &= 2\,\mathrm{softplus}\big[f_\theta(\Delta_t)\big]. \end{aligned}} \quad (11)$$

The same f_θ aggregates every tracking objective, so PPO never consults e_g, a hand-set success threshold, or a component coefficient. The global factor 2 is a fixed reward scale, shared across skills.

Because softplus is 1-Lipschitz, the critic bound passes to the signal that PPO actually receives. On the already normalized differential (4),

$$\big|r^{\mathrm{DARE}}(\Delta) - r^{\mathrm{DARE}}(0)\big| \le 2\big|f_\theta(\Delta) - f_\theta(0)\big| \le 2\|\Delta\|_2. \quad (12)$$

Nearby residuals therefore produce nearby rewards, in the same units in which the critic is trained.

Training alternates rollout collection, differential construction, critic updates, and PPO updates (Algorithm 1). After each control step the synchronized residual is formed with (4), converted to r_t^{DARE}, and written to a replay buffer together with the on-policy batch. Critic updates draw an equal mixture of current and replay differentials, so q_π is

Algorithm 1 DARE Training

Require: reference motions $\mathcal{M}$, policy π_φ, value V_ω, critic f_θ, replay buffer $\mathcal{B}$

for iteration = $1, 2, \ldots$ **do**
  Collect a synchronized rollout with π_φ
  Construct Δ_t using (4)
  Compute r_t^{DARE} using (11)
  Add rollout differentials to $\mathcal{B}$
  Sample current and replay differentials
  Update f_θ by minimizing (5)
  Estimate returns and advantages from r_t^{DARE}
  Update π_φ and V_ω with PPO
end for

not a single-iteration snapshot. The critic minimizes (5); spectral parameterization already enforces (10). The policy builds returns from (11) and applies the standard PPO surrogate. DARE learns only an instantaneous scalar reward. Long-horizon credit assignment remains the responsibility of discounting, GAE, and PPO.

The critic f_θ scores a single residual; the value function V_ω still predicts the discounted return of (11). Keeping those roles separate leaves long-horizon credit assignment with PPO, which is already the standard interface for reference-conditioned control. The change is concentrated in how the differential is represented and scored, not in a new policy class.

TABLE I

Frozen DARE reward-learning configuration. All entries are identical for every motion.

Component	Setting
Semantic factors	7 (root pos./rot., body pos./rot., root lin./ang. vel., DoF vel.)
Direct-sum width	146 per factor (1022 total)
Shared trunk	SN MLP 1022 → 512, ReLU
Output	SN logit, affine bias
First-layer bias	yes, zero-initialized
Critic optimizer	SGD, 2.5×10^{-4}, wd 10^{-4}
Logit regularizer	10^{-2}
Replay	buffer 2×10^5, 1000 samples/batch
Reward scale	2
Normalization	running per-coordinate scale
Gradient penalty	0
Spectral maps	all linear layers

V. Experiments

A. Experimental Setup

All methods control a 28-DoF simulated humanoid with PD joint targets at 30 Hz; the simulator runs at 120 Hz in Isaac Lab [26], [27]. The character model follows the simulated humanoid used by DeepMimic and ADD [28]–[30]. Episodes are initialized at a random reference phase. DeepMimic, ADD, and DARE use pose termination with a unit threshold. AMP follows its standard protocol without pose termination, because a distribution-matching prior is not synchronized to a reference frame. Early termination on undesired ground contact is enabled for every method. Each policy is trained for 2000 iterations with 8192 parallel environments and 32 control steps per iteration, for a budget of 5.243×10^8 environment samples. Actor and value networks are two-layer MLPs with 1024 units. Learned-reward methods share the PPO clip 0.2, GAE $\lambda = 0.95$, discount $\gamma = 0.99$, and SGD family; DARE additionally shares one critic architecture and one reward-learning configuration across motions.

The evaluation uses six skills that span distinct dynamical regimes: cyclic ground rotation (Roll, 2.00 s), long-horizon object interaction (Climb, 9.97 s), aerial inversion (Backflip, 1.75 s), prone locomotion (Crawl, 2.93 s), recumbent recovery (Get-Up, 3.03 s), and a ballistic kick (Spin Kick, 1.28 s). Motion files, reference lengths, object layouts, and termination conditions belong to the task definition; they are not treated as shared hyperparameters. Baselines use the training settings published with each method. We report position tracking error in meters and DoF-velocity tracking error in radians per second, the two quantities used by prior humanoid tracking studies. AMP, DeepMimic, and ADD numbers in Table II are those reported by Zhang *et al.* for the same clips and metrics [17]. Every DARE run uses the frozen configuration in Table I, identical for all six skills. Each method is trained with five seeds and evaluated on 4096 test episodes; we report mean ± std.

B. Cross-Motion Tracking Performance

Table II asks whether one frozen reward-learning configuration can serve skills whose contacts, horizons, and velocity profiles differ. DARE matches or improves published tracking accuracy on five of the six clips, and remains within the DeepMimic range on Roll position, without restoring a weighted tracking sum and without choosing a critic coefficient per skill. Short, highly constrained recoveries (Crawl, Get-Up, Spin Kick) already have little room under ADD; DARE stays in the same band while using the configuration of Table I. Longer and more acrobatic clips leave more aggregation work for the critic. There the equalized differential improves both position and DoF-velocity error relative to AMP, and it improves velocity tracking relative to DeepMimic, which still depends on handcrafted kernels.

The six rows are not six copies of the same control problem. Roll is a cyclic ground skill in which root travel and body rotation must stay in phase. Backflip inverts the character and then demands a precise landing velocity. Crawl keeps the torso near the ground, so small orientation errors become large position errors. Get-Up starts from a recumbent pose and has to discover a contact sequence that restores stance. Spin Kick is short and ballistic: a mistimed root rotation cannot be repaired later in the clip. Across that range, the same DARE critic, optimizer, replay buffer, and reward scale train every row of the table. That is the empirical claim: reuse of the reward-learning configuration, rather than a new champion number on every metric.

Accompanying video shows synchronized reference and policy frames for each skill. Those frames are qualitative

TABLE II

Tracking performance on six heterogeneous skills. DARE uses one frozen reward-learning configuration on every row. AMP, DeepMimic, and ADD are published results on the same clips and metrics [17]. Mean ± std over five seeds. Lower is better.

		Position [m] ↓				DoF vel. [rad/s] ↓			
Skill	Len. (s)	AMP	DeepMimic	ADD	DARE	AMP	DeepMimic	ADD	DARE
Roll	2.00	.141±.031	.115±.132	.152±.005	.118±.012	1.576±.318	.994±.051	1.330±.101	1.012±.041
Climb	9.97	.185±.021	.066±.004	.046±.031	.042±.008	1.190±.012	.567±.015	.374±.071	.352±.018
Backflip	1.75	.267±.015	.111±.054	.062±.001	.058±.004	2.243±.113	1.103±.024	.878±.013	.812±.024
Crawl	2.93	.050±.006	.027±.000	.028±.002	.027±.003	.646±.089	.430±.006	.283±.002	.271±.012
Get-Up	3.03	.096±.018	.023±.001	.022±.001	.021±.002	.838±.029	.433±.008	.325±.005	.308±.009
Spin Kick	1.28	.064±.010	.078±.062	.025±.000	.024±.003	1.453±.327	1.222±.233	.774±.007	.731±.016

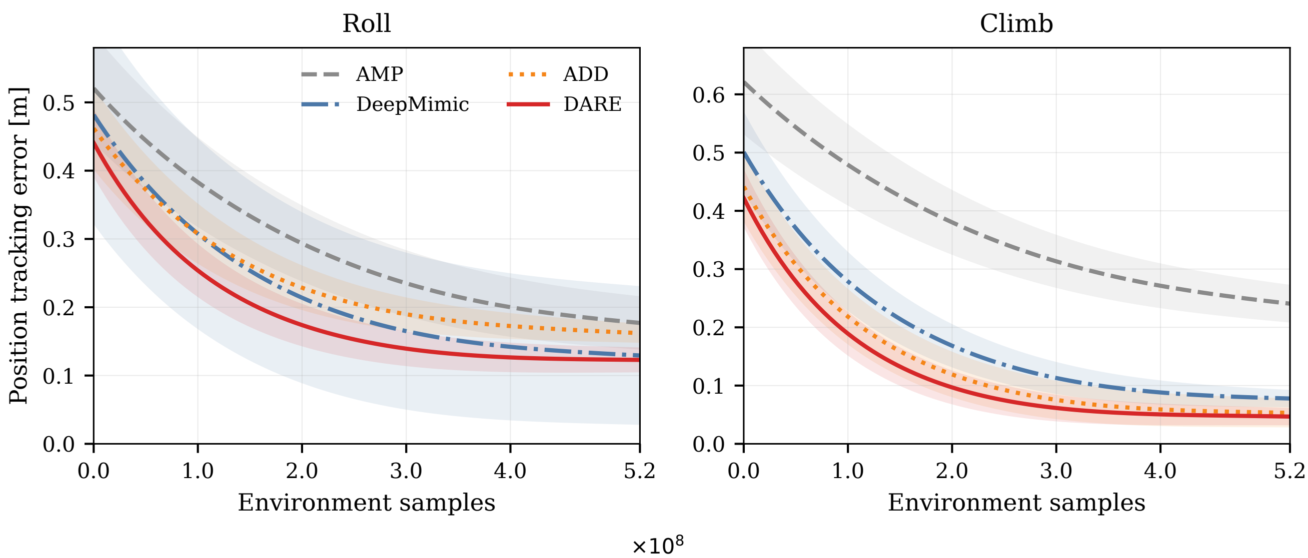

Fig. 1. Position tracking error versus environment samples on two representative regimes. Curves are seed means; shaded regions are standard deviations. Final six-skill numbers are in Table II.

support for completion; Table II is the quantitative comparison.

C. Component Ablation

Table III isolates the two design choices on Roll and Climb, chosen as representative short-cycle and long-horizon regimes. The full six-skill result is already in Table II; the ablation is not repeated on every clip. All three variants share the actor, PPO, training budget, differential definition, and network capacity. *Flat+Full SN* replaces the direct-sum layer by a spectrally normalized dense map of the same output width, so the first nonlinearity already mixes every coordinate. *Factorized+GP* keeps the direct-sum layer but replaces spectral gain control by a gradient penalty of coefficient 2, frozen before evaluation and shared by both skills. DARE keeps both factorization and spectral parameterization, with no gradient penalty.

The largest degradation comes from flattening the first layer. Body-pose and joint coordinates then outnumber root position by more than an order of magnitude, and the Lipschitz trunk inherits that imbalance. Position error rises from 0.118 m to 0.168 m on Roll and from 0.042 m to 0.091 m on the long-horizon clip; velocity error rises with it. The shared trunk is still nonlinear and still 1-Lipschitz. What it no longer receives is an equalized view of each physical residual, so the prices it learns follow coordinate cardinality rather than contact, balance, or timing. That is why the first layer must preserve the blocks of (4): later mixing is then about residual geometry, not about how many numbers a feature happens to occupy.

Factorized+GP lands between the two. Semantic channels are intact, yet sensitivity is controlled by a scalar coefficient rather than by the maps themselves. The same coefficient is usable on both representative skills, which is already more reusable than a per-clip penalty, but the resulting errors remain higher than DARE. Nothing in that architecture gives a three-dimensional root residual and a high-dimensional pose residual equal standing before mixing. Structured gain control and equalization belong together: factorization decides *what* enters the aggregator, and the bound (10) decides *how steeply* that aggregator may respond.

D. Representative Learning Curves

Figure 1 plots position tracking error against environment samples for the same two representative regimes. All four methods are shown under a matched training budget; solid

TABLE III

ABLATION ON TWO REPRESENTATIVE SKILLS. MEAN ± STD OVER FIVE SEEDS. LOWER IS BETTER.

Variant	Position [m] ↓ Roll	Position [m] ↓ Climb	DoF vel. [rad/s] ↓ Roll	DoF vel. [rad/s] ↓ Climb
Flat+Full SN	.168±.021	.091±.019	1.284±.087	.612±.041
Factorized+GP	.139±.015	.061±.011	1.156±.052	.448±.027
DARE	.118±.012	.042±.008	1.012±.041	.352±.018

curves are seed means and shaded regions are standard deviations. On Roll, DeepMimic attains a slightly lower mean than DARE, but with a much wider seed band; DARE sits next to that mean and continues to improve after AMP has flattened. The long-horizon clip takes longer for every method, as expected from a multi-second contact sequence. There DARE and ADD separate from AMP, and DARE edges the published ADD plateau while keeping a comparable schedule. DoF-velocity results are already in Table II and are not repeated as a second figure.

VI. Conclusion

Humanoid imitation is a multi-objective tracking problem. Root motion, body pose, and velocity have to be satisfied together, and their coupling is strongest on the contact-rich skills that make reward engineering painful. Manual rewards turn that aggregation into per-motion engineering: each new clip invites a new choice of weights, scales, and regularizers. DARE instead learns a nonlinear scalar reward from the synchronized motion differential, and it keeps each semantic error intact until a shared critic is allowed to mix them.

Because perfect imitation occupies a unique zero differential, DARE bounds the learned score—and the PPO reward—relative to that anchor. Nearby residuals cannot produce disproportionate reward jumps, while the shared trunk remains free to represent the combinations required by different motions. Ablating the first layer shows that this reuse depends on keeping each semantic residual intact until that trunk is allowed to mix them. The same architecture, optimizer, and reward-learning configuration therefore train Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick, with position errors from $0.021\,\mathrm{m}$ to $0.118\,\mathrm{m}$ and DoF-velocity errors from $0.271\,\mathrm{rad/s}$ to $1.012\,\mathrm{rad/s}$. Accompanying video shows synchronized reference and policy frames for each skill.

References

[1] X. B. Peng, P. Abbeel, S. Levine, and M. van de Panne, “DeepMimic: Example-guided deep reinforcement learning of physics-based character skills,” *ACM Trans. Graph.*, vol. 37, no. 4, 2018.

[2] J. Won, D. Gopinath, and J. Hodgins, “A scalable approach to control diverse behaviors for physically simulated characters,” *ACM Trans. Graph.*, vol. 39, no. 4, 2020.

[3] T. Wang, Y. Guo, M. Shugrina, and S. Fidler, “UniCon: Universal neural controller for physics-based character motion,” arXiv:2011.15119, 2020.

[4] Z. Luo, J. Cao, A. W. Winkler, K. Kitani, and W. Xu, “Perpetual humanoid control for real-time simulated avatars,” in *Proc. ICCV*, 2023.

[5] Z. Luo, J. Cao, J. Merel, A. Winkler, J. Huang, K. M. Kitani, and W. Xu, “Universal humanoid motion representations for physics-based control,” in *Proc. ICLR*, 2024.

[6] C. Tessler, Y. Guo, O. Nabati, G. Chechik, and X. B. Peng, “MaskedMimic: Unified physics-based character control through masked motion inpainting,” *ACM Trans. Graph.*, 2024.

[7] J. Merel, Y. Tassa, D. TB, S. Srinivasan, J. Lemmon, Z. Wang, G. Wayne, and N. Heess, “Learning human behaviors from motion capture by adversarial imitation,” arXiv:1707.02201, 2017.

[8] N. Heess, D. TB, S. Sriram, J. Lemmon, J. Merel, G. Wayne, Y. Tassa, T. Erez, Z. Wang, S. M. A. Eslami, M. Riedmiller, and D. Silver, “Emergence of locomotion behaviours in rich environments,” arXiv:1707.02286, 2017.

[9] X. B. Peng, A. Kanazawa, J. Malik, P. Abbeel, S. Levine, and S. Fidler, “SFV: Reinforcement learning of physical skills from videos,” *ACM Trans. Graph.*, vol. 37, no. 6, 2018.

[10] K. Bergamin, S. Clavet, D. Holden, and J. R. Forbes, “DReCon: Data-driven responsive control of physics-based characters,” *ACM Trans. Graph.*, vol. 38, no. 4, 2019.

[11] S. Park, H. Ryu, S. Lee, S. Lee, and J. Lee, “Learning predict-and-simulate policies from unorganized and simulated motion data,” *ACM Trans. Graph.*, vol. 38, no. 4, 2019.

[12] J. Ho and S. Ermon, “Generative adversarial imitation learning,” in *Advances in Neural Information Processing Systems*, 2016.

[13] X. B. Peng, Z. Ma, P. Abbeel, S. Levine, and A. Kanazawa, “AMP: Adversarial motion priors for stylized physics-based character control,” *ACM Trans. Graph.*, vol. 40, no. 4, 2021.

[14] X. B. Peng, Y. Guo, L. Halper, S. Levine, and S. Fidler, “ASE: Large-scale reusable adversarial skill embeddings for physically simulated characters,” *ACM Trans. Graph.*, vol. 41, no. 4, 2022.

[15] J. Fu, K. Luo, and S. Levine, “Learning robust rewards with adversarial inverse reinforcement learning,” in *Proc. ICLR*, 2018.

[16] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. ICML*, 2017.

[17] Z. Zhang, S. Bashkirov, D. Yang, Y. Shi, M. Taylor, and X. B. Peng, "Physics-based motion imitation with adversarial differential discriminators," in *SIGGRAPH Asia Conf. Papers*, 2025.

[18] T. Salimans and D. P. Kingma, "Weight normalization: A simple reparameterization to accelerate training of deep neural networks," in *Advances in Neural Information Processing Systems*, 2016.

[19] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. Courville, "Improved training of Wasserstein GANs," in *Advances in Neural Information Processing Systems*, 2017.

[20] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *Proc. ICLR*, 2018.

[21] O. Sener and V. Koltun, "Multi-task learning as multi-objective optimization," in *Advances in Neural Information Processing Systems*, 2018.

[22] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. CVPR*, 2018.

[23] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, "Gradient surgery for multi-task learning," in *Advances in Neural Information Processing Systems*, 2020.

[24] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[25] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *Proc. ICLR*, 2016.

[26] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac Gym: High performance GPU-based physics simulation for robot learning," in *Proc. NeurIPS Datasets and Benchmarks*, 2021.

[27] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, and A. Garg, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robot. Autom. Lett.*, vol. 8, no. 6, pp. 3740–3747, 2023.

[28] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2012.

[29] N. Chentanez, M. Müller, M. Macklin, V. Makoviychuk, and S. Jeschke, "Physics-based motion capture imitation with deep reinforcement learning," in *Proc. ACM SIGGRAPH Conf. Motion, Interaction and Games*, 2018.

[30] S. Lee, M. Park, K. Lee, and J. Lee, "Scalable muscle-actuated human simulation and control," *ACM Trans. Graph.*, vol. 38, no. 4, 2019.